EE426 / 626 Microprocessor Applications

Bare Metal vs CMSIS vs HAL



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HAL – Hardware Abstration Layer (STMicroelectronics Specific)

```
# include “stm32f4xx_hal.h”

============================================================
HAL — ST's abstraction layer, config passed through a struct
============================================================

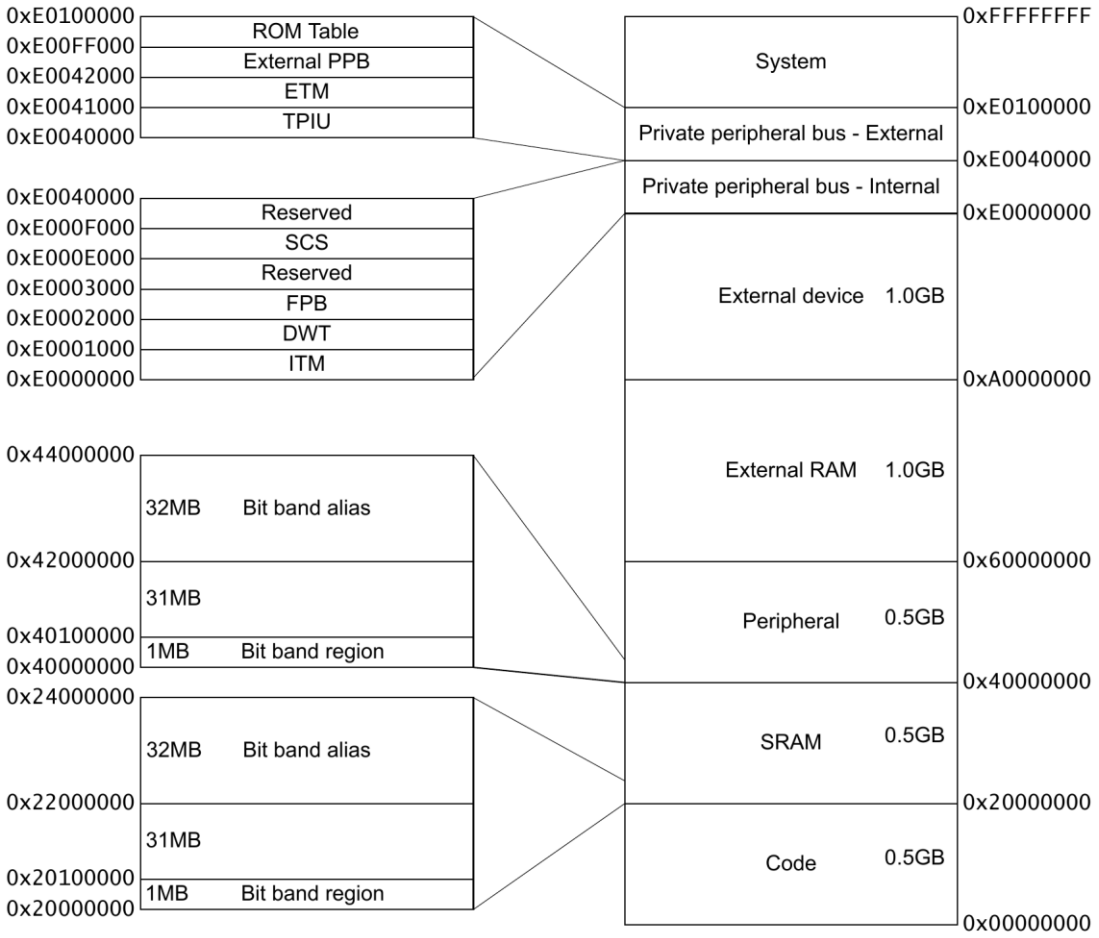
  GPIO_InitTypeDef GPIO_InitStruct = {0};

  GPIO_InitStruct.Pin   = GPIO_PIN_5;
  GPIO_InitStruct.Mode  = GPIO_MODE_OUTPUT_PP;
  GPIO_InitStruct.Pull  = GPIO_NOPULL;
  GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_LOW;

  HAL_GPIO_Init(GPIOA, &GPIO_InitStruct);
```



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STM32F446RE
Memory Map

0x4002 1C00 - 0x4002 1FFF	GPIOH	AHB1	*Section 7.4.11: GPIO register map on page 191*
0x4002 1800 - 0x4002 1BFF	GPIOG		
0x4002 1400 - 0x4002 17FF	GPIOF		
0x4002 1000 - 0x4002 13FF	GPIOE		
0x4002 0C00 - 0x4002 0FFF	GPIOD		
0x4002 0800 - 0x4002 0BFF	GPIOC		
0x4002 0400 - 0x4002 07FF	GPIOB		
0x4002 0000 - 0x4002 03FF	GPIOA		



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Bare Metal – Direct access to peripheral registers

```
============================================================
 BARE METAL — raw pointer dereference to the register address
============================================================

  #define GPIOA_BASE   0x40020000UL
  #define GPIOA_MODER  (*(volatile uint32_t *)(GPIOA_BASE + 0x00UL))

  /* Pin 5 -> MODER bits [11:10]. Set to 0b01 = general purpose output */
  GPIOA_MODER |=  (1U << 10);   /* MODER5 bit 0 -> 1 */
  GPIOA_MODER &= ~(1U << 11);   /* MODER5 bit 1 -> 0 */
```



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CMSIS – Cortex Microcontroller Software Interface Standard

```
typedef struct {
  __IO uint32_t MODER;    /*!< GPIO port mode register,                Address offset: 0x00 */
  __IO uint32_t OTYPER;   /*!< GPIO port output type register,         Address offset: 0x04 */
  __IO uint32_t OSPEEDR;  /*!< GPIO port output speed register,        Address offset: 0x08 */
  __IO uint32_t PUPDR;    /*!< GPIO port pull-up/pull-down register,   Address offset: 0x0C */
  __IO uint32_t IDR;      /*!< GPIO port input data register,          Address offset: 0x10 */
  __IO uint32_t ODR;      /*!< GPIO port output data register,         Address offset: 0x14 */
  __IO uint32_t BSRR;     /*!< GPIO port bit set/reset register,       Address offset: 0x1A */
  __IO uint32_t LCKR;     /*!< GPIO port configuration lock register, Address offset: 0x1C */
  __IO uint32_t AFR[2];   /*!< GPIO alternate function registers,      Address offset: 0x20-0x24 */
  __IO uint32_t BRR;      /*!< GPIO bit reset register,                 Address offset: 0x28 */
} GPIO_TypeDef;

/* --- The base addresses: bus -> peripheral, built up in layers --- */

#define PERIPH_BASE         0x40000000UL
#define AHB2PERIPH_BASE      (PERIPH_BASE + 0x08000000UL)
#define GPIOA_BASE            (AHB2PERIPH_BASE + 0x00000000UL)


#define GPIOA                ((GPIO_TypeDef *) GPIOA_BASE)
```



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CMSIS – Cortex Microcontroller Software Interface Standard

```
# include "stm32f446xc.h"

============================================================
  CMSIS – device-header struct overlay of the same address
============================================================

  GPIOA->MODER |=  (1U << 10);
  GPIOA->MODER &= ~(1U << 11);
```

You are strongly encouraged to use CMSIS for this course (not HAL) !!

